Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. If two jointly distributed random variables are independent, are they uncorrelated also?

1 / 1 point

- (•) Yes: they are always uncorrelated also.
- ( ) Sometimes: they may be correlated or they may be uncorrelated.
- ( ) No: they are never uncorrelated also.

Correct
Yes. Independent random variables are always uncorrelated: independence is a strong condition that states that there is no relationship between the random variables.

2. If two jointly distributed random variables are uncorrelated, are they independent also?

1 / 1 point

- ( ) Yes: they are always independent also.
- (•) Sometimes: they may be independent or they may not be independent.
- ( ) No: they are never independent also.

Correct
Yes. They may be independent or they may not be independent. There is no general rule.

3. If $f_{X|Y}(x|y) = f_X(x)$, what can we say about random variables X and Y? (Select all that must apply.)

1 / 1 point

- [x] X and Y are independent.

Correct
Yes. Knowledge of the value of $Y = y$ does not change the pdf of X, so X and Y must be independent. Also, this can only happen when $f_{X,Y}(x,y) = f_X(x)f_Y(y)$, which is the condition for independence.

- [x] X and Y are uncorrelated.

Correct
Yes. The indicated condition implies that the variables are independent, which requires that they also be uncorrelated.

- [ ] X and Y have a joint Gaussian distribution.
- [x] The conditional pdf $f_{Y|X}(y|x) = f_Y(y)$.

Correct
Yes. This is a consequence of X and Y being independent. You can also show this using Bayes' rule.